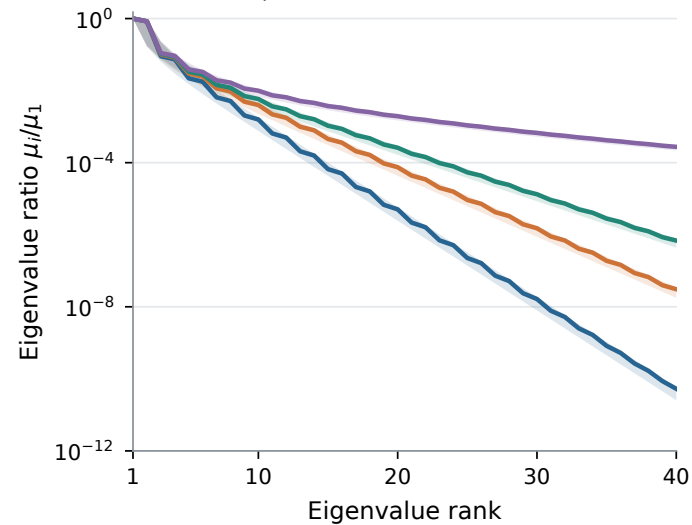


A Gamma sets relative learning rates

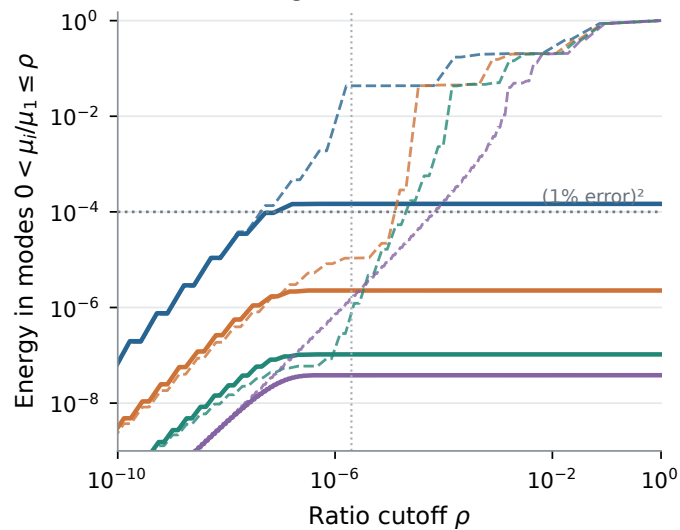
Lines: finite spectrum Bands: lemma intervals



$\gamma=8$ $\gamma=16$
 $\gamma=12$ $\gamma=64$

B Adam leaves slow-mode residual

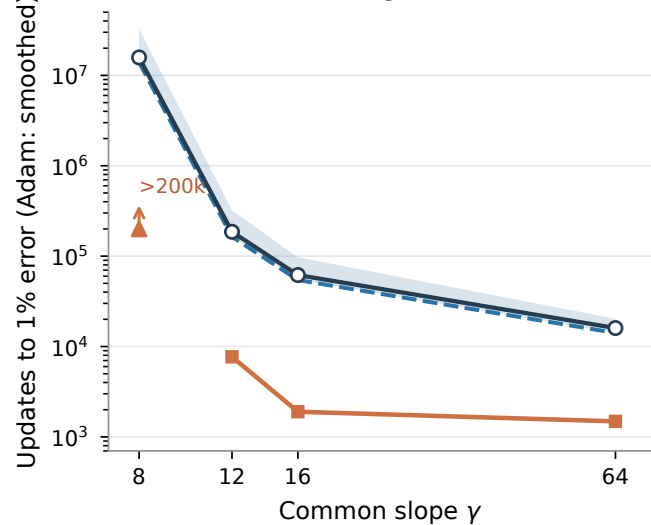
Dashed: initial target Solid: Adam at 200k



$\gamma=8$ $\gamma=16$
 $\gamma=12$ $\gamma=64$

C Error lower bound forces delay

Adam: first loss-EMA crossing, half-life 100



GD time interval Executed GD
 Theorem: necessary GD time Adam first EMA hit
 GD spectrum forecast